

Approximate linear programming for decentralized policy iteration in cooperative multi-agent Markov decision processes

Vidhi Sonkar, Biplav Oraon, Basab Ghosh

October 1, 2026

Indian Institute of Science, Bangalore

Problem Statement

- ▶ Cooperative MA-MDP with m agents, common objective
- ▶ Challenges:
 - ▶ Large state space (COD)
 - ▶ Exponential action space
- ▶ Decentralized PI reduces action complexity
- ▶ Still needs **exact value function** (costly)
- ▶ **Goal:** Scalable PI via **approximation**

Multi-agent MDP

- ▶ MDP: (S, U, P, g)
- ▶ Joint action: $u = (u_1, \dots, u_m)$, $U = \prod_{i=1}^m U_i$
- ▶ Policy: $\mu = (\mu_1, \dots, \mu_m)$
- ▶ Value:

$$J^\mu(x) = \mathbb{E} \left[\sum \alpha^k g(x_k, u_k) \right]$$

- ▶ **Issue:** $|U|$ grows exponentially in m

Base Paper Contributions [1]

- ▶ Introduce **ADPI**: ALP-based decentralized policy iteration for cooperative MA-MDPs.
- ▶ Applicable to **finite** and **infinite horizon** settings.
- ▶ Use **function approximation** to avoid costly exact value computation.
- ▶ Provide **theoretical guarantees**
- ▶ **Scalable** to large state-action spaces and many agents.
- ▶ **Empirically faster** convergence with competitive or better performance

Algorithm 1 (Decentralized Policy Improvement (DPIm))

Input: Agent i , updated policies $\tilde{\mu}_{1:i-1}$, base policies $\mu_{i+1:m}$, value function J

Update rule:

$$\tilde{\mu}_i(x) \in \arg \min_{u_i \in U_i(x)} \sum_{y \in S} p_{xy}(\tilde{\mu}_1, \dots, \tilde{\mu}_{i-1}, u_i, \mu_{i+1}, \dots, \mu_m) [g_{xy}(\cdot) + \alpha J(y)]$$

Output: Updated policy $\tilde{\mu}_i$

Algorithm 2 (CACFN)

Input: Stage k , policy π , next-stage cost $J_{k+1,\tilde{\pi}}^{\text{ALP}}$

Compute: Solve the following ALP

$$\begin{aligned} J_{k,\tilde{\pi}}^{\text{ALP}} &= \max_{r \in \mathbb{R}^d} c^\top \Phi r \\ \text{s.t. } \Phi r(x) &\leq \sum_{y \in S} p_{xy}(\tilde{\mu}_1, \dots, \tilde{\mu}_m) [g_{xy}(\cdot) + J_{k+1,\tilde{\pi}}^{\text{ALP}}(y)], \quad \forall x \in S \end{aligned}$$

Output: Approximate cost $J_{k,\tilde{\pi}}^{\text{ALP}} = \Phi r$

Algorithm 3 : DPI using ALP : FH

Input: Horizon N , agents m , initial policy π

Initialize: $J_{N,\pi}^{\text{ALP}} = g_N$

For $k = N - 1, \dots, 0$:

► **Repeat until convergence:**

- **Policy Improvement:** For $i = 1, \dots, m$, update $\tilde{\mu}_i$ using DPI (Alg. 1)
- **Policy Evaluation:** Compute $J_{k,\tilde{\pi}}^{\text{ALP}}$ using ALP (Alg. 2)
- Update policy: $\pi \leftarrow \tilde{\pi}$

Output: Approximate optimal policy π

Algorithm 4 (Approximate Cost Function (IH))

Input: Policy μ

Compute: Solve

$$r_\mu = \arg \max_{r \in \mathbb{R}^d} c^\top \Phi r \quad \text{s.t. } T_\mu(\Phi r) \geq \Phi r$$

Output: $J_\mu^{\text{ALP}} = \Phi r_\mu$

Algorithm 5 (DPI using ALP for IH CO-MA-MDP)

Input: Initial policy μ^0 , discount α

Initialize: Compute $J_{\mu^0}^{\text{ALP}}$ (Alg. 4)

For $t = 0, 1, \dots$:

- ▶ **Policy Improvement:** For $i = 1, \dots, m$, update $\tilde{\mu}_i$ using DPIIm (Alg. 1)
- ▶ **Policy Update:** $\mu^{t+1} \leftarrow \tilde{\mu}$
- ▶ **Policy Evaluation:** Compute $J_{\mu^{t+1}}^{\text{ALP}}$ (Alg. 4)

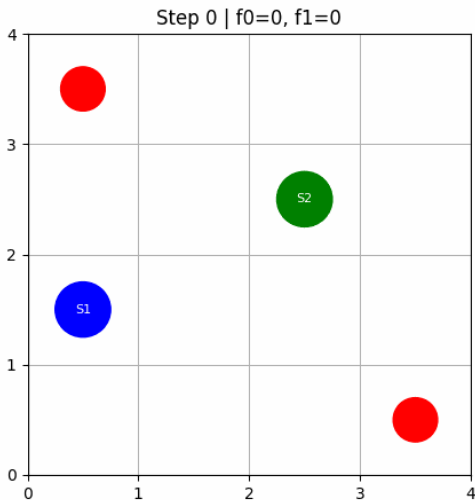
Output: Final policy μ^T

Experiment

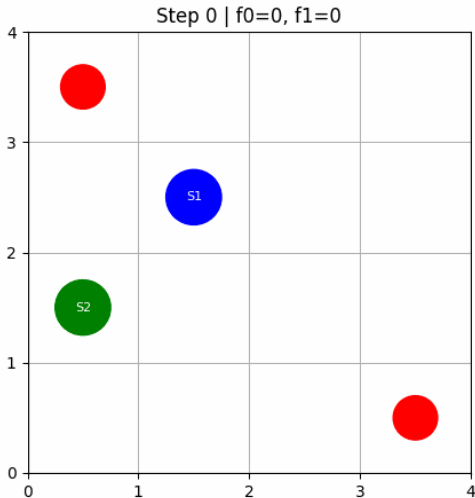
We implemented Algorithms 1-5 of Lakshmi et al. (2025) [1]. We performed the Spider-Flies experiment in the following settings:

- ▶ 4×4 grid world with 2 spiders (agents) and 2 fixed flies (goal states)
- ▶ State Space: 256 joint states representing positions of both agents
- ▶ Actions: up, down, left, right
- ▶ Cost: +1 step cost, +2 any agent out of grid, +2 collision of agents
- ▶ Terminal Cost: 0

Finite Horizon, Algorithm 3, $N = 10$



Infinite Horizon, Algorithm 5, $\alpha = 0.95$



Our Contributions

- ▶ Provided empirical results to show performance of the algorithms in the Base Paper
- ▶ Showed reduction of Infinite Horizon setting to Stochastic Shortest Path (SSP) problem under some conditions
- ▶ Provided theoretical analysis and empirical results for the same
- ▶ Performed stability analysis of SSP setting

Experimental Results (Base Paper)

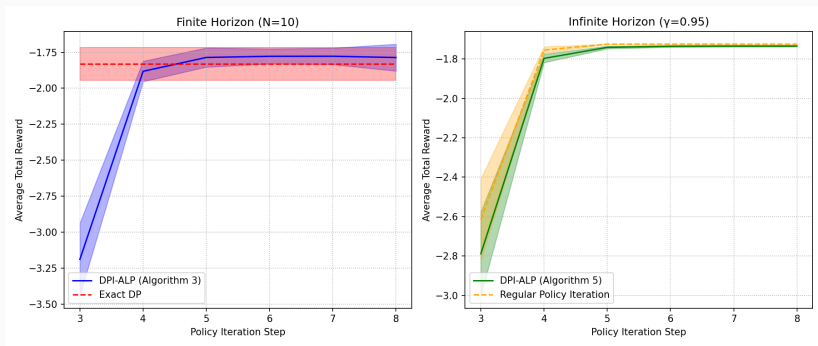


Figure 1: Convergence of Algorithm 3 and Algorithm 5, for Spider-Flies experiment

SSP Setup

To realize SSP, we:

- ▶ Use proper policies
- ▶ Set discount factor as 1

We do Decentralized Policy Iteration using Algorithm 5 of the Base Paper and compare it against the exact Policy Iteration in the SSP setting

Experimental Results SSP

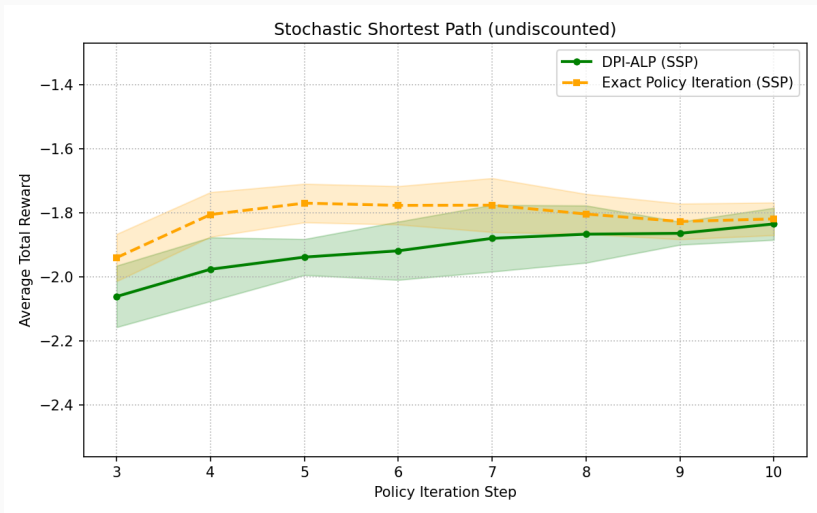


Figure 2: SSP with proper policies

Notations

We consider a finite-state SSP with:

State Space: $\mathcal{S} = \mathcal{S}' \cup \{t\}$, where $\mathcal{S}'$ is the set of non-terminal states, t is the terminal state

Action Space: $\mathcal{A}(x)$ = set of feasible actions in state $x \in \mathcal{S}$

Stationary policy: μ

Transition probabilities: $p(y \mid x, a)$ = probability of transitioning from state $x \in \mathcal{S}$ to state $y \in \mathcal{S}$ taking action $a \in \mathcal{A}(x)$

Immediate Cost: $g(x, y, a)$ = cost for transition from state $x \in \mathcal{S}$ to state $y \in \mathcal{S}$ taking action $a \in \mathcal{A}(x)$

Discount factor: $\alpha \in (0, 1)$

Notations (continued)

Boundary conditions:

$$p(t \mid t, a) = 1 \quad \forall a \in \mathcal{A}(t)$$

$$g(t, t, a) = 0 \quad \forall a \in \mathcal{A}(t)$$

One-stage Cost: We define one-stage cost at a state $x \in \mathcal{S}'$ as:

$$g_\mu(x) := \sum_{y \in \mathcal{S}'} p(y \mid x, \mu(x))g(x, y, \mu(x)) + p(t \mid x, \mu(x))g(x, t, \mu(x)) \quad \forall x \in \mathcal{S}'$$

Transition matrix P_μ for any vector J on $\mathcal{S}'$ and $x \in \mathcal{S}'$:

$$(P_\mu J)(x) := \sum_{y \in \mathcal{S}'} p(y \mid x, \mu(x))J(y)$$

Feasible Sets

The Feasible set of IH-ALP is given as:

$$\mathcal{F}_\alpha = \{r \in \mathbb{R}^d \mid \Phi' r \leq g_\mu + \alpha P_\mu \Phi' r\}$$

Feasible set of SSP-ALP is:

$$\mathcal{F}_{\text{ssp}} = \{r \in \mathbb{R}^d \mid \Phi' r \leq g_\mu + P_\mu \Phi' r\}$$

Here Φ' is the feature matrix on $\mathcal{S}'$. Both LPs maximize $c^\top \Phi' r$ with $c \geq 0$.

Containment of Feasible Sets

Lemma 1. For $\alpha \in (0, 1)$, $\mathcal{F}_\alpha \subseteq \mathcal{F}_{\text{ssp}}$

Proof. Since μ is proper, $I - P_\mu$ is invertible and each element of $(I - P_\mu)^{-1}$ is non-negative.

For $\alpha \in (0, 1)$, $I - \alpha P_\mu$ is also invertible and $(I - \alpha P_\mu)^{-1}$ has non-negative elements.

We define matrix M_α as:

$$\begin{aligned} M_\alpha &= (I - P_\mu)(I - \alpha P_\mu)^{-1} \\ &= ((I - \alpha P_\mu) + (\alpha P_\mu - P_\mu))(I - \alpha P_\mu)^{-1} \\ &= I - (1 - \alpha)P_\mu(I - \alpha P_\mu)^{-1} \\ &\leq I \end{aligned} \tag{1}$$

Proof Continued

Let $r \in \mathcal{F}_\alpha$, i.e., $(I - \alpha P_\mu)\Phi' r \leq g_\mu$.

Multiplying both sides by M_α we obtain:

$$(I - P_\mu)\Phi' r = M_\alpha(I - \alpha P_\mu)\Phi' r \leq M_\alpha g_\mu \leq g_\mu$$

Hence, $r \in \mathcal{F}_{\text{ssp}}$



Inner limit – convergence of feasible sets

Lemma 2. For every $\bar{r} \in \mathcal{F}_{\text{ssp}}$ there exists a sequence $r_\alpha \in \mathcal{F}_\alpha$ with $r_\alpha \rightarrow \bar{r}$ as $\alpha \rightarrow 1^-$.

Proof. Let $\bar{r} \in \mathcal{F}_{\text{ssp}}$, so we have $(I - P_\mu)\Phi'\bar{r} \leq g_\mu$. Computing violation at $\bar{r}$:

$$\begin{aligned}(I - \alpha P_\mu)\Phi'\bar{r} - g_\mu &= (I - P_\mu)\Phi'\bar{r} - g_\mu + (1 - \alpha)P_\mu\Phi'\bar{r} \\ &\leq (1 - \alpha)P_\mu\Phi'\bar{r}\end{aligned}\tag{2}$$

Hence, the positive part satisfies

$$\|((I - \alpha P_\mu)\Phi'\bar{r} - g_\mu)^+\|_\infty \leq (1 - \alpha) \|P_\mu\Phi'\bar{r}\|_\infty$$

Proof Continued

The **Hoffman bound** states that for a full-column-rank matrix A ,

$$\text{dist}(r_0, \{r \mid Ar \leq b\}) \leq \frac{\|(Ar_0 - b)^+\|_\infty}{\sigma_{\min}(A)},$$

where $\sigma_{\min}(A) > 0$ is the smallest singular value of A .

Assuming Φ' has full column rank, and under proper policy μ , $(I - \alpha P_\mu)$ is invertible, therefore, $(I - \alpha P_\mu)\Phi'$ has full column rank.

$$\text{dist}(\bar{r}, \mathcal{F}_\alpha) \leq \frac{(1 - \alpha)\|P_\mu \Phi' \bar{r}\|_\infty}{\sigma_{\min}((I - \alpha P_\mu)\Phi')}$$

Under bounded costs, we get $\text{dist}(\bar{r}, \mathcal{F}_\alpha) \rightarrow 0$ as $\alpha \rightarrow 1^-$. We can choose r_α such that $\|r_\alpha - \bar{r}\| \rightarrow 0$ and $r_\alpha \in \mathcal{F}_\alpha$. □

Value Convergence

Lemma 3. Let $v_\alpha = \max_{r \in \mathcal{F}_\alpha} c^\top \Phi' r$ and $v_{\text{ssp}} = \max_{r \in \mathcal{F}_{\text{ssp}}} c^\top \Phi' r$. Then
 $\lim_{\alpha \rightarrow 1^-} v_\alpha = v_{\text{ssp}}$

Proof. From Lemma 1, $\mathcal{F}_\alpha \subseteq \mathcal{F}_{\text{ssp}}$, so $v_\alpha \leq v_{\text{ssp}}$.

Let $\bar{r}^*$ be an optimal solution to SSP-ALP (exists by compactness of the set). By Lemma 2, there exist $r_\alpha \in \mathcal{F}_\alpha$ with $r_\alpha \rightarrow \bar{r}^*$ as $\alpha \rightarrow 1^-$.

Then $c^\top \Phi' r_\alpha \leq v_\alpha$. Taking limit inferior on both sides as $\alpha \rightarrow 1^-$, we get: $\liminf_{\alpha \rightarrow 1^-} c^\top \Phi' r_\alpha \leq \liminf_{\alpha \rightarrow 1^-} v_\alpha$. Since $r_\alpha \rightarrow \bar{r}^*$ and $c^\top \Phi'$ is continuous, $\liminf_{\alpha \rightarrow 1^-} c^\top \Phi' r_\alpha = v_{\text{ssp}}$. Thus $v_{\text{ssp}} \leq \liminf_{\alpha \rightarrow 1^-} v_\alpha$.

Combining both inequalities, we have:

$$\liminf_{\alpha \rightarrow 1^-} v_\alpha \leq \limsup_{\alpha \rightarrow 1^-} v_\alpha \leq v_{\text{ssp}} \leq \liminf_{\alpha \rightarrow 1^-} v_\alpha$$

Therefore, $\lim_{\alpha \rightarrow 1^-} v_\alpha = v_{\text{ssp}}$

Optimizer Convergence

Theorem 3. Assume the SSP-ALP has a unique optimal solution r^{ssp} (Φ' has full column rank and $c > 0$). Let $r^*(\alpha)$ be any optimal solution of the discounted ALP. Then

$$\lim_{\alpha \rightarrow 1^-} \Phi' r^*(\alpha) = \Phi' r^{\text{ssp}}.$$

Proof. All feasible sets lie in a compact set (bounded by $(I - P_\mu)^{-1}g_\mu$), so $\{r^*(\alpha)\}$ has a convergent subsequence. Take any convergent subsequence $r^*(\alpha_n) \rightarrow \hat{r}$.

For each n , $(I - \alpha_n Q_\mu)\Phi' r^*(\alpha_n) \leq \bar{g}_\mu$. Taking limits and using continuity gives $(I - Q_\mu)\Phi' \hat{r} \leq \bar{g}_\mu$, so $\hat{r} \in \mathcal{F}_{\text{ssp}}$.

By continuity of $c^\top \Phi'$ and value convergence, $c^\top \Phi' \hat{r} = \lim_n c^\top \Phi' r^*(\alpha_n) = \lim_n v_{\alpha_n} = v_{\text{ssp}}$. Thus $\hat{r}$ is optimal for the SSP-ALP.

Proof Continued

The SSP-ALP has a unique optimal solution, hence $\hat{r} = r^{\text{ssp}}$.

Therefore, every convergent subsequence converges to r^{ssp} . If the whole sequence did not converge to r^{ssp} , there would exist a subsequence staying away from it, which would itself have a convergent sub-subsequence (by compactness) whose limit would differ from r^{ssp} – a contradiction.

Hence, $r^*(\alpha) \rightarrow r^{\text{ssp}}$.

Finally, continuity of Φ' yields $\Phi' r^*(\alpha) \rightarrow \Phi' r^{\text{ssp}}$. □

Goal: SSP Analogue of Theorem 2 [1]

Objective: Stability guarantees in SSP settings

Key Question:

- ▶ Does approximate policy iteration remain stable without discounting?
- ▶ How does error propagate when α is removed?

Key Idea:

$$\frac{1}{1 - \alpha} \longrightarrow H = \mathbb{E}[\tau]$$

Main Result

Theorem (SSP Analogue of Theorem 2 [1])

$$J^{\mu_{t+1}}(x) \leq J^{\mu_t}(x) + H\beta_t$$

$$\beta_t = \|J^{\mu_t} - J_{\mu_t}^{ALP}\|_{\infty}$$

Assumptions

► **Proper Policies:**

$$\mathbb{P}_x^\mu(\tau < \infty) = 1, \quad \mathbb{E}_x^\mu[\tau] < \infty$$

► **Bounded Costs:**

$$0 \leq g(x, u, y) \leq g_{\max}$$

► **Approximation Error:**

$$\|J^\mu - J_\mu^{\text{ALP}}\|_\infty \leq \beta$$

► **Bounded Hitting Time:**

$$\sup_{x \in S} \mathbb{E}_x^{\mu_{t+1}}[\tau] \leq H$$

Step 1: Bellman Update

We begin by evaluating one step under the updated policy:

$$(T_{\tilde{\mu}}J^{\mu})(x) = \sum_y P(y|x, \tilde{\mu}(x)) [g(x, \tilde{\mu}(x), y) + J^{\mu}(y)]$$

Using the approximation error bound:

$$J^{\mu}(y) \leq J_{\mu}^{\text{ALP}}(y) + \beta$$

Substituting into the Bellman operator:

$$T_{\tilde{\mu}}J^{\mu}(x) \leq \sum_y P(\cdot) \left[g + J_{\mu}^{\text{ALP}}(y) \right] + \beta$$

Step 2: Greedy Policy Improvement

We rewrite the Bellman operator using the approximate value:

$$T_{\tilde{\mu}} J_{\mu}^{\text{ALP}}(x) = \sum_y P(\cdot) \left[g + J_{\mu}^{\text{ALP}}(y) \right]$$

Since the policy is greedy w.r.t. J_{μ}^{ALP} :

$$T_{\tilde{\mu}} J_{\mu}^{\text{ALP}}(x) \leq T_{\mu} J_{\mu}^{\text{ALP}}(x)$$

Combining:

$$T_{\tilde{\mu}} J^{\mu}(x) \leq T_{\mu} J_{\mu}^{\text{ALP}}(x) + \beta$$

Step 3: ALP Lower Bound

From ALP feasibility, the approximation is a lower bound:

$$J_{\mu}^{\text{ALP}}(x) \leq J^{\mu}(x)$$

Applying the Bellman operator:

$$T_{\mu} J_{\mu}^{\text{ALP}}(x) \leq T_{\mu} J^{\mu}(x) = J^{\mu}(x)$$

Thus, we obtain the key inequality:

$$T_{\tilde{\mu}} J^{\mu}(x) \leq J^{\mu}(x) + \beta$$

Step 4: Error Propagation

Applying the Bellman operator repeatedly:

$$T_{\tilde{\mu}}^k J^{\mu}(x) \leq J^{\mu}(x) + k\beta$$

Observation:

- ▶ Error grows linearly with steps
- ▶ No discount factor to shrink it

Step 5: Stopping Time Argument

Let τ be the hitting time of the terminal state.

Evaluating at stopping time:

$$T_{\tilde{\mu}}^{\tau} J^{\mu}(x) \leq J^{\mu}(x) + \tau \beta$$

This replaces the infinite horizon with termination.

Step 6: Taking Expectation

Taking expectation over trajectories:

$$J^{\tilde{\mu}}(x) = \mathbb{E}_x^{\tilde{\mu}} [T_{\tilde{\mu}}^{\tau} J^{\mu}(x)]$$

Using linearity of expectation:

$$J^{\tilde{\mu}}(x) \leq J^{\mu}(x) + \beta \cdot \mathbb{E}_x^{\tilde{\mu}}[\tau]$$

Final Result: SSP Stability

Using bounded hitting time:

$$\mathbb{E}_x^{\tilde{\mu}}[\tau] \leq H$$

We obtain:

$$J^{\tilde{\mu}}(x) \leq J^{\mu}(x) + H\beta$$

Conclusion:

- ▶ Error is controlled by expected trajectory length
- ▶ Algorithm remains stable in SSP

References

-  Lakshmi Mandal, et al. (2025). *Approximate linear programming for decentralized policy iteration in cooperative multi-agent MDPs*, *Systems & Control Letters*, 196.
-  Dimitri P. Bertsekas (2020). *Dynamic Programming and Optimal Control Volume I*.
-  Roger A. Horn, Charles R. Johnson (2012). *Matrix Analysis*. Cambridge University Press.
-  D. P. De Farias, B. Van Roy (2002). *The Linear Programming Approach to Approx DP*. 36